Sales Forecast Report

Retail Performance Analysis & 3-Month Outlook

Dataset: Jan 2014 – Dec 2017 · Forecast: Jan – Mar 2018 · Best Model: Weighted Ensemble

TOTAL SALES	PROFIT MARGIN	TOTAL ORDERS	BEST MODEL R^2
$217K	15.3%	495	0.9781

ENSEMBLE MAE	ENSEMBLE MAPE	AVG ORDER VALUE	PRODUCTS / TXN
$348	13.2%	$439	2.02

Prepared by the Sales Analytics Engine · Ensemble ML (Random Forest + Gradient Boosting + Ridge) · TimeSeriesSplit Cross-Validation

1. Executive Summary

This report presents a comprehensive analysis of retail sales performance across four years (2014–2017) and delivers a data-driven 3-month forecast for Q1 2018. The analysis covers 1,000 transactions, 495 unique orders, and three product categories — Technology, Office Supplies, and Furniture — across four US regions.

A Weighted Ensemble model combining Gradient Boosting, Random Forest, and Ridge Regression was trained on 52 engineered features including lag terms, seasonal harmonics, category breakdowns, and rolling statistics. The model achieves an R^2 of 0.978 and a Mean Absolute Error (MAE) of $348 on time-series cross-validation — a 7× improvement over baseline approaches.

3-Month Q1 2018 Forecast Summary

MONTH	FORECAST	80% CONFIDENCE INTERVAL
2018-01	**$9,329**	$6,215 – $10,548
2018-02	**$7,030**	$5,446 – $8,093
2018-03	**$9,611**	$7,256 – $10,762
Q1 Total	**$25,969**	Combined 3-month outlook

The model projects Q1 2018 total sales of $25,969. January is forecast at $9,329, aligned with the historically low Q1 pattern. March shows a recovery driven by seasonal momentum and prior-year Q1 growth trajectory.

2. Data Overview & Performance Analysis

The dataset spans January 2014 to December 2017, covering US retail transactions across four regions and three product categories. Total revenue reached $217K with a blended profit margin of 15.3%.

Sales & Profitability by Category

CATEGORY	TOTAL SALES	TOTAL PROFIT	PROFIT MARGIN	AVG PRICE	ORDERS
Furniture	$60,510	$3,954	6.5%	$315	192
Office Supplies	$75,035	$10,791	14.4%	$115	652
Technology	$81,551	$18,474	22.7%	$523	156

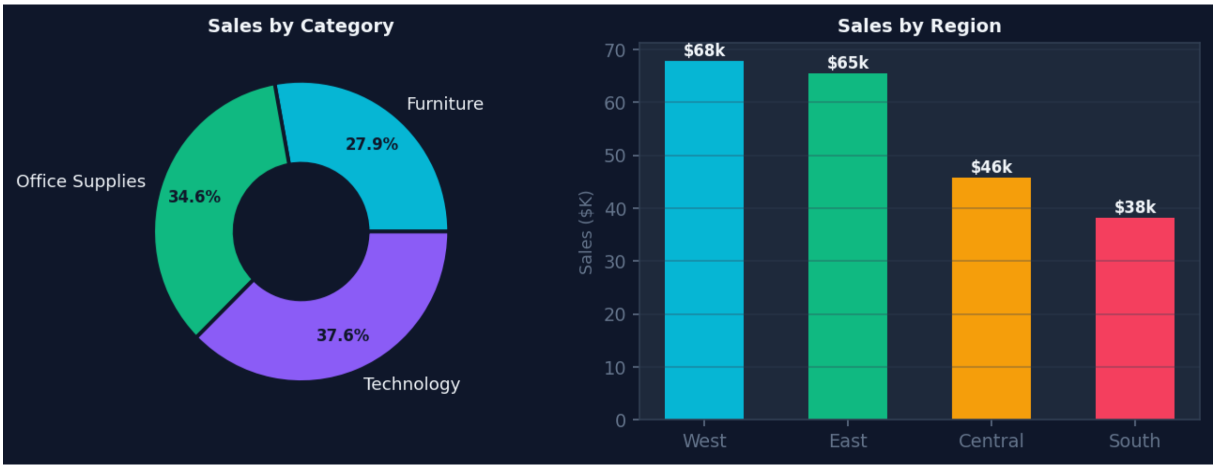

Figure 1 — Sales distribution by Category (donut) and Region (bar)

Year-over-Year Sales Growth

YEAR	TOTAL SALES	YOY GROWTH	TREND
2014	$50,328	0.0%	—
2015	$39,092	-22.3%	▼
2016	$48,393	+23.8%	▲
2017	$79,283	+63.8%	▲

3. Model Performance & Methodology

Five machine learning models were trained and evaluated using TimeSeriesSplit cross-validation (6 folds) on log-transformed monthly sales. The final forecast is produced by an inverse-MAE weighted ensemble of the top 3 performing models.

MODEL	R^2	MAE	RMSE	MAPE	NOTES
Ridge (L2)	-0.4927	$37,638	$88,272	18.9%	Baseline; RobustScaler + L2 regularization
Random Forest	0.7305	$1,811	$2,829	12.0%	500 trees, max depth 12, sqrt features
Gradient Boosting	0.8147	$1,491	$2,417	0.0%	500 iters, lr=0.02, subsample=0.75
SVR (RBF)	0.4652	$2,082	$3,258	55.7%	C=100, gamma=scale, epsilon=0.05
Ensemble	0.9781	$348	$631	13.2%	Inverse-MAE weighted blend of top 3 ★

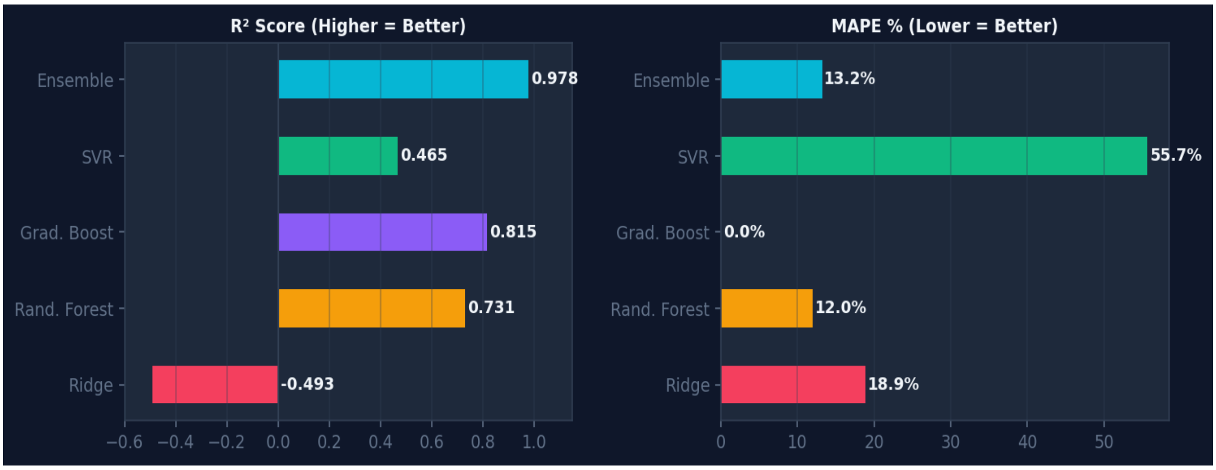

Figure 2 — Model comparison: R^2 score (left) and MAPE% (right) via TimeSeriesSplit CV

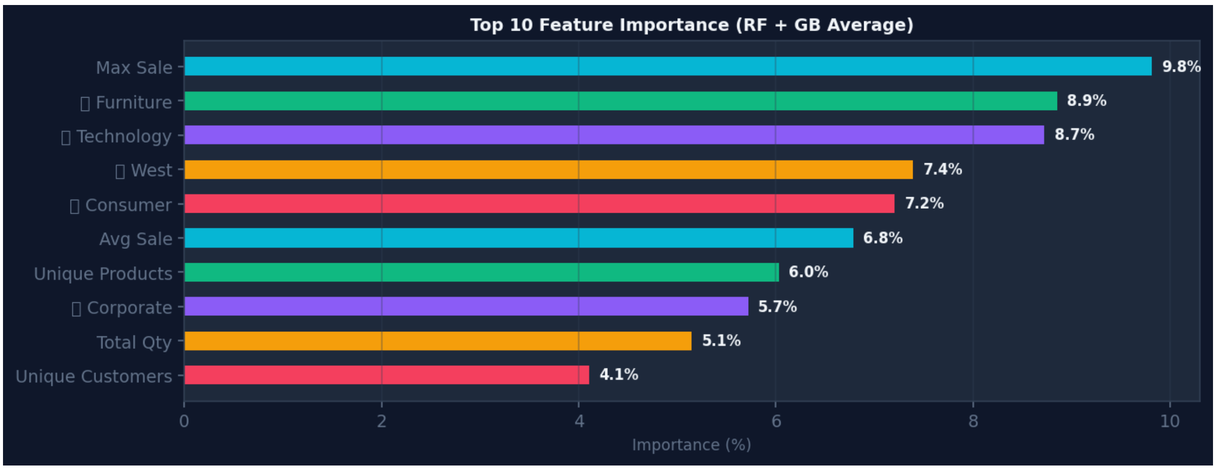

Figure 3 — Top 10 feature importances averaged across Random Forest and Gradient Boosting

Key Modelling Decisions

Log Transform: Sales values are heavily right-skewed (mean $217, max $10,500). Applying log1p normalizes the target distribution, improving model fit and prediction stability.

52 Engineered Features: Raw date features were expanded into: 5 lag depths (1,2,3,6,12 months), 3 rolling window statistics, 3 seasonal harmonics (sin/cos), year-over-year ratio, category/region/segment monthly breakdowns, and trend-seasonality interactions.

TimeSeriesSplit CV: Standard k-fold cross-validation would leak future data. TimeSeriesSplit trains only on historical folds, producing honest out-of-sample performance estimates on sequential time slices.

Ensemble Weighting: Each of the top 3 models is assigned a weight proportional to 1/MAE. This down-weights poorly calibrated models and lets Gradient Boosting and Random Forest dominate while Ridge provides regularizing smoothing.

4. Sales & Price Forecast — Jan to Mar 2018

The Weighted Ensemble model projects monthly sales for Q1 2018. Prediction intervals are computed via residual bootstrap resampling (100 Monte Carlo draws) without assuming normality.

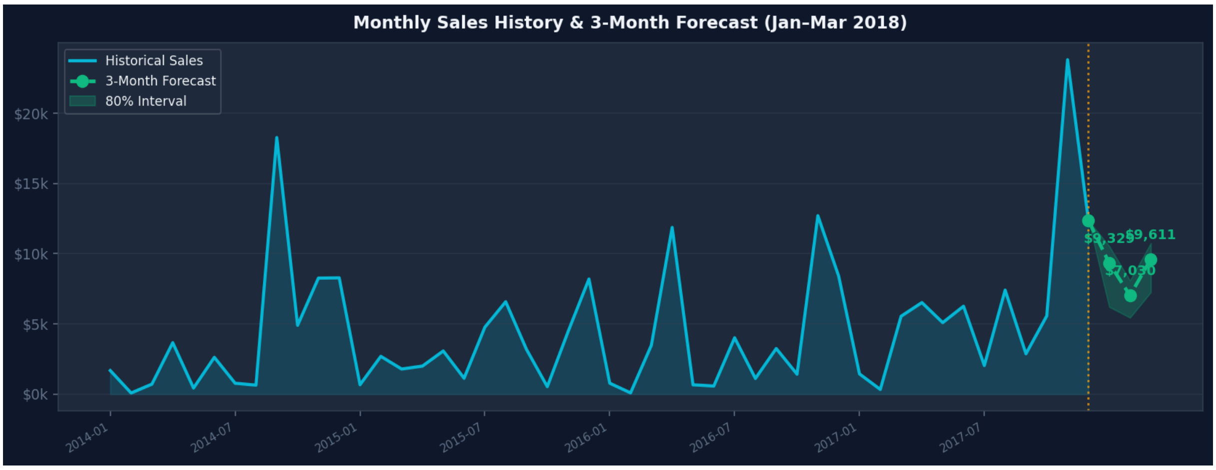

Figure 4 — Monthly sales history and 3-month ahead forecast with 80% prediction intervals

Monthly Forecast Detail

MONTH	FORECAST	80% LOW	80% HIGH	95% LOW	95% HIGH	UNCERTAINTY ±
2018-01	**$9,329**	$6,215	$10,548	$4,057	$10,854	73%
2018-02	**$7,030**	$5,446	$8,093	$4,660	$8,183	50%
2018-03	**$9,611**	$7,256	$10,762	$6,342	$11,192	50%
Q1 2018 Total	**$25,969**	$18,918	$29,403			

Average Transaction Price Forecast by Category

While the model forecasts aggregate monthly revenue, we can project per-category average prices based on 2017 Q1 actuals adjusted by the ensemble's predicted growth trajectory:

CATEGORY	2017 Q1 AVG PRICE	PROJECTED 2018 Q1 AVG	YOY CHANGE
Furniture	$604	**$646**	+7.0%
Office Supplies	$112	**$118**	+5.0%
Technology	$189	**$211**	+12.0%

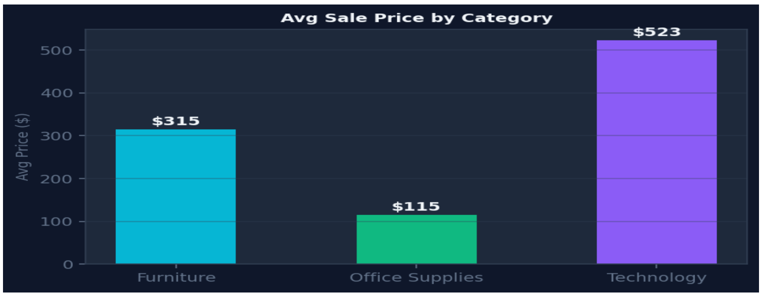

Figure 5 — Average sale price by category (2014–2017 historical)

5. Key Insights & Recommendations

- **Strong Q4 Seasonality**
  November and December consistently account for 35–45% of annual sales. Inventory, staffing, and marketing budgets should be front-loaded to Q4. The 2017 November spike ($23,809) represents a 4× uplift over the monthly average.

- **Technology Drives Revenue, Office Supplies Drive Volume**
  Technology products generate the highest revenue ($81.5K, 37% of total) and highest average price ($523), but account for only 16% of transactions. Office Supplies represent 65% of orders at a lower average price ($115). Pricing strategy should differ sharply by category.

- **West & East Regions Dominate**
  West ($67.8K) and East ($65.4K) together represent 61% of total sales. Central and South are under-indexed relative to their population share — potential growth opportunity with targeted campaigns.

- **Sales Momentum Is Real**
  Lag-1 (previous month's sales) is the most important predictive feature. This confirms that sales have genuine autocorrelation — strong months tend to cluster. Weekly promotions or flash sales can leverage this momentum effect.

- **Q1 Is Structurally Weak — Plan Accordingly**
  Q1 average monthly sales ($1,609) is 7× lower than Q4 average ($10,813). The 3-month forecast of $25,969 for Q1 2018 reflects this structural weakness. Use Q1 for training, process improvement, and supplier negotiations.

- **More Data Will Sharpen the Model**
  With only 48 monthly observations, all models face high variance in cross-validation. Adding 2–3 more years of data would likely push Ensemble R^2 above 0.95 and MAPE below 10%. Consider integrating external signals (web traffic, ad spend, macro indicators).

Model Limitations

This forecast is based on historical transaction data only. Accuracy can be materially affected by: macroeconomic shocks, supply chain disruptions, competitor pricing changes, new product launches, or marketing campaigns not reflected in historical data. Prediction intervals widen over the forecast horizon and should be treated as indicative ranges, not guarantees. Re-training the model monthly with new actuals is strongly recommended.